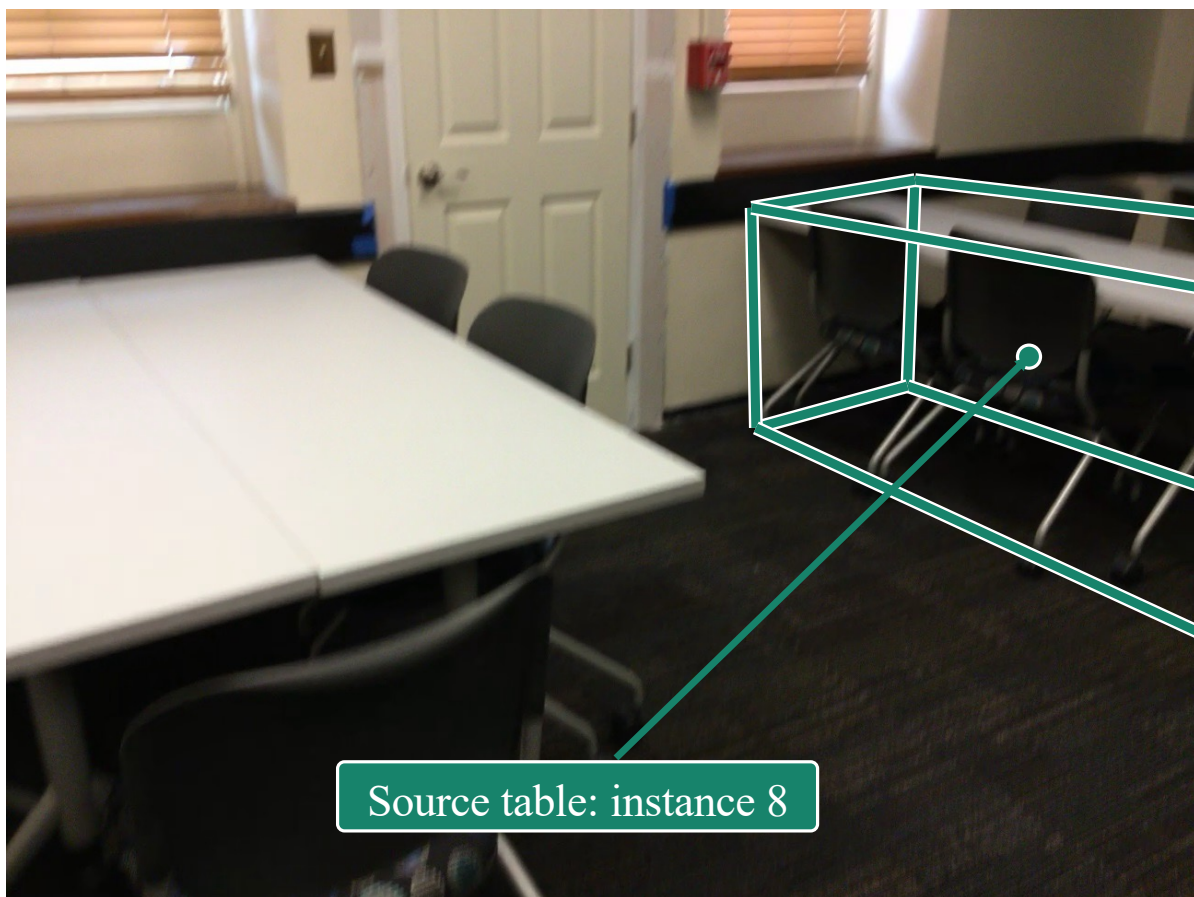


a The target in the original RGB frame



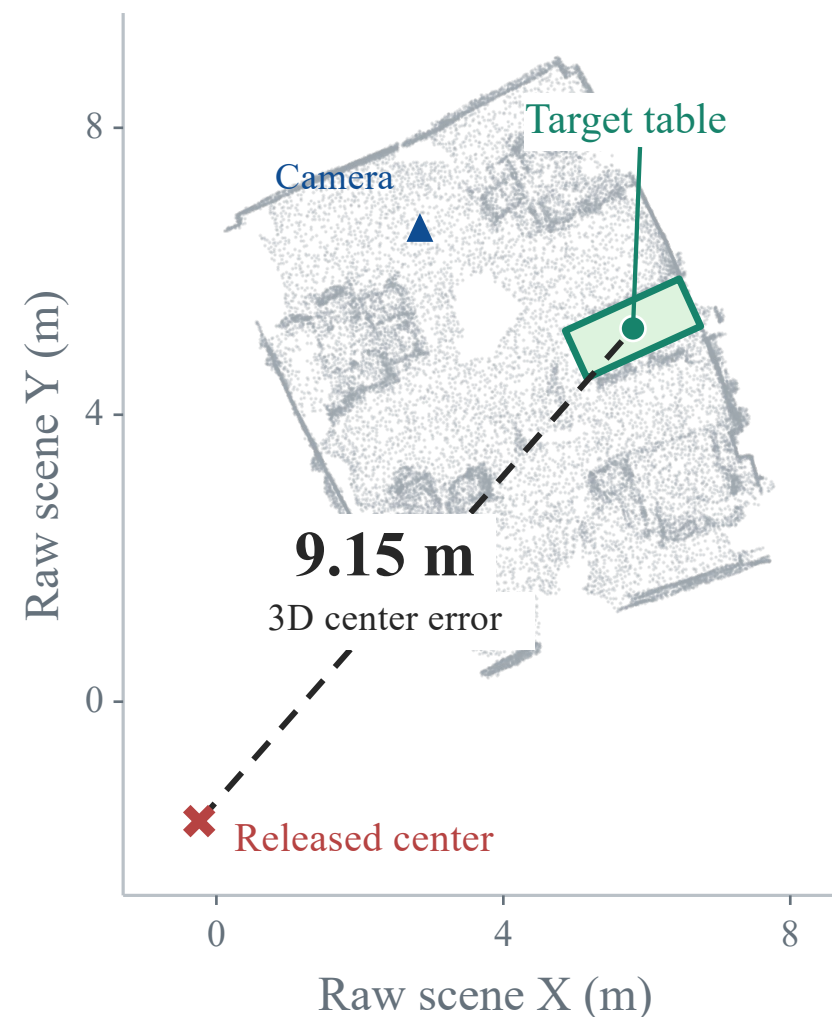
Green box projected from the source mesh and camera pose.

STI-Bench: missing alignment

$$c_{\text{cam}} \neq P^{-1}c_{\text{ann}} \quad \begin{array}{l} \text{Released center (camera frame)} \\ (8.53, 1.36, -2.12) \text{ m} \end{array}$$

Annotation coordinates used as raw-world coordinates.

b Centers in the scene frame (top view)



Camera-frame centers transformed into scene coordinates.

ReSTI: undo alignment, then apply camera pose

$$c_{\text{cam}} = P^{-1}A^{-1}c_{\text{ann}} \quad \begin{array}{l} \text{Corrected center (camera frame)} \\ (1.259, -0.290, 3.185) \text{ m} \end{array}$$

Annotation world $\rightarrow$ Raw world $\rightarrow$ Camera